



Grade 11 Geography

6. Mass Movements

Learning Guide

1. What is Mass Movement?

- **Mass movement** is also called **mass wasting**.
- It is the **downward movement of weathered material under the force of gravity**.
- The movement may be:
 - **Slow** → e.g. soil creep
 - **Rapid** → e.g. landslides and mudflows
- Material may move downslope by:
 - **Creeping**
 - **Sliding**
 - **Flowing**
 - **Falling**

 **Key idea:** Gravity is the main force responsible for moving material downslope.

2. What Causes Mass Movement?

- Mass movement occurs when the force of **gravity becomes greater than the resistance of the slope**.
- When this happens, the slope can no longer remain stable and **slope failure** occurs.

Slope Failure

- Happens when **gravitational pull exceeds the internal friction** holding slope material together.
- The slope then collapses or begins to move downslope.

3. Factors Influencing Mass Movement

Several natural conditions determine whether a slope will remain stable or fail.

1. Slope Gradient

- A **steeper slope** allows material to move more quickly.
- Therefore, steep slopes are generally more prone to mass movement.

2. Rock Structure

- Weaknesses in rocks make movement easier.
- Examples include:
 - **Joints**
 - **Bedding planes**
- Material may slide along these weaknesses.

3. Vegetation

- Vegetation helps to **stabilise slopes**.
- Plant roots anchor soil.
- Vegetation also removes some water from the soil.
- This reduces the likelihood of slope failure.

4. Soil Type

- Some soils move more easily when wet.
- **Saturated clay soils** are especially likely to move downslope.

5. Rainfall

- Rain adds water and **weight to the slope**.
- This increases instability and may trigger movement.

6. Earthquakes

- Earthquake vibrations can:
 - Loosen rocks.
 - Dislodge material.
 - Trigger movement on unstable slopes.

4. Human Activities that Trigger Mass Movements

Human activities can weaken slopes and increase the chance of slope failure.

Building on Steep Slopes

- Buildings add extra **weight** to the slope.

Road Construction and Blasting

- Cutting and blasting into slopes can **weaken the slope structure**.

Quarrying

- Quarrying near the **base of a slope** may remove support.
- This can cause the slope above to collapse.

Deforestation and Overgrazing

- Remove vegetation.
- This removes the roots that normally **anchor the soil**.

Poor Farming Practices

- Ploughing **down the slope** loosens soil.
- Loose material is more easily moved.

Dumping Debris

- Adds loose and unstable material to slopes.
- This can increase the risk of movement.

5. Types of Mass Movement

From **slow to fast**:

1. Soil creep
2. Solifluction
3. Earthflow
4. Mudflow
5. Slumping
6. Landslide
7. Rockfall

6. Soil Creep

- **Soil creep** is the **slowest form of mass movement**.
- Movement is approximately **1 cm per year**.
- It involves very slow downslope movement of soil particles.

Cause

- Expansion and contraction of soil grains.
- This may occur under:
 - **Dry conditions**
 - **Cold conditions**

Where It Occurs

- **Crest slopes**
- **Talus slopes**

Feature Formed

- **Terracettes** develop.
- These are small **step-like features** on the slope.

Signs of Soil Creep

- Bent fences.
- Tilting trees or poles.
- Cracked walls.

 **Key idea:** Soil creep is extremely slow, but its effects become visible over time.

7. Solifluction

- **Solifluction** is the **slow flow of water-saturated soil** downslope.

Cold Areas

- The upper soil layer **thaws**.
- The subsoil remains **frozen**.
- Water cannot easily move downward, so the saturated upper soil slowly flows over the frozen layer.

Arid Areas

- Solifluction may also occur after rainfall.

Characteristics

- Forms **tongue-like lobes of soil**.
 - Usually occurs on **talus slopes**.
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8. Earthflow

- **Earthflow** is the movement of **wet clay** down a slope.
- It is more rapid than soil creep and solifluction.

Characteristics

- Forms **tongue-shaped lobes**.
- May continue for:
 - Hours.
 - Or even years.
- Occurs mainly on:
 - **Talus**
 - **Pediment**

 **Key idea:** Earthflow involves wet clay moving downslope in a lobe-like mass.

9. Mudflow

- A **mudflow** is the **rapid movement of saturated fine particles**.
- It is particularly associated with **arid areas after heavy rainfall**.

Characteristics

- Fine material becomes saturated with water.
- It moves rapidly downslope.
- It forms **mud streams in definite channels**.

Where It Occurs

- **Talus slopes**
- **Pediments**

 **Key idea:** Mudflow = fine material + lots of water + rapid downslope movement.

10. Slumping

- **Slumping** is also called a **rotational landslide**.
- A block of land slides downslope along a **curved surface**.

Characteristics

- The moving block may **tilt backwards**.
- Common on **steep slopes**.
- Often leaves behind a **curved scar**.

 **Key idea:** Slumping involves rotation along a curved surface.

11. Landslides

- A **landslide** is the **sudden and rapid movement of material** downslope.
- Common where **bedding planes are parallel to the slope angle**.

Characteristics

- Material slides along a relatively flat or **planar surface**.

- Leaves behind a **planar rupture surface**.
- Common in:
 - **Jointed rocks**
 - **Layered rocks**

 **Key idea:** *Landslide = rapid sliding along a relatively straight surface.*

12. Rockfalls

- **Rockfalls** occur on **cliffs or very steep rock faces**.
- They are triggered when weathering weakens rock.

Causes

- **Physical weathering**
- **Chemical weathering**

Process

1. Rock becomes weakened.
2. Pieces detach from the cliff.
3. Rocks fall **vertically downward**.
4. Fallen rocks collect at the base.

Result

- Material accumulates on the **talus slope**.

Common Locations

- Mountains.
- Sea cliffs.

 **Key idea:** *Rockfall = weathered rock breaks from a steep face and falls vertically.*

13. Comparison of the Main Types of Mass Movement

Type	Speed	Main Conditions	Slope Area
Soil creep	Very slow	Dry/cold conditions, subtle tilting	Crest, talus
Solifluction	Slow	Frozen or water-saturated soil	Talus
Earthflow	Moderate	Wet clay, forms lobes	Talus, pediment
Mudflow	Rapid	Arid area + heavy rain	Talus, pediment
Slumping	Fast	Curved sliding surface, steep slope	Steep slopes
Landslide	Fast	Planar slide, layered rock	Slope face
Rockfall	Fast	Weathered cliffs	Talus/base

14. How to Tell the Types Apart

Soil Creep

Very slow + leaning trees/fences + terracettes.

Solifluction

Saturated soil + slow flow + cold/wet conditions.

Earthflow

Wet clay + tongue-shaped lobe.

Mudflow

Fine saturated material + heavy rain + channelised flow.

Slumping

Block rotates along a **curved** surface.

Landslide

Rapid movement along a **planar/straight** surface.

Rockfall

Rocks break from cliffs and **fall vertically**.



15. Final Exam Summary

- **Mass movement / mass wasting** is the downward movement of weathered material due to **gravity**.
 - It occurs when:
Gravity > slope resistance → slope failure.
 - Factors influencing mass movement include:
 - Slope gradient.
 - Rock structure.
 - Vegetation.
 - Soil type.
 - Rainfall.
 - Earthquakes.
 - Human activities can increase instability through:
 - Building.
 - Road construction and blasting.
 - Quarrying.
 - Deforestation.
 - Overgrazing.
 - Poor farming.
 - Dumping debris.
 - **Soil creep** is the slowest type.
 - **Solifluction** is slow movement of saturated soil.
 - **Earthflow** involves wet clay forming lobes.
 - **Mudflow** is rapid movement of saturated fine material.
 - **Slumping** occurs along a curved surface.
 - **Landslides** move rapidly along a planar surface.
 - **Rockfalls** involve rocks falling vertically from steep cliffs.
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